

UNIVERSITÉ PARIS-DAUPHINE

DEEP LEARNING PROJECT

Emotion-Conditioned Music Generation

*Study and Implementation of a Conditional Variational Autoencoder
(CVAE-LSTM) on the EMOPIA Dataset*

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Emotion-Conditioned Music Generation

1 Introduction

Music is fundamentally an emotional language. Long before analyzing its harmonic, rhythmic, or melodic structure, listeners immediately perceive an intention, an atmosphere, or a feeling. However, musical emotion is inherently subjective. It depends not only on sound waves, but also on the listener’s culture, musical background, and psychological state at the time of listening. A melody perceived as melancholy by one person may feel soothing to another.

This intrinsic subjectivity presents a major hurdle for artificial intelligence. Most generative models produce music that, while technically sound, lacks direction and intention. The resulting pieces often sound random, unable to convey genuine human emotion.

The primary objective of this project is to generate music that adheres to a specific emotional intent. Rather than creating an AI that outputs arbitrary notes, we aim to build a system capable of composing "on demand," guided by explicit emotional prompts (e.g., joy, anger, sadness, calmness).

To teach a machine this subjectivity, emotion must first be mapped into a measurable framework. To this end, we utilize the EMOPIA dataset, which annotates piano compositions using Russell’s circumplex model.

We first evaluate various Variational Autoencoder (VAE) architectures to identify the most effective baseline for music generation[cite: 1]. We then transition to a Conditional Variational Autoencoder (CVAE) combined with Long Short-Term Memory networks (LSTM) to generate distinct compositions tailored to specific target emotions[cite: 1].

2 Data Description and Preprocessing

2.1 The EMOPIA Dataset

This project relies on the EMOPIA (Emotion in Piano Performance) dataset[cite: 1]. It consists of solo piano recordings annotated according to Russell’s circumplex model of emotion[cite: 1].

2.1.1 Russell’s Circumplex Model (1980)

Developed by psychologist James Russell, this model frames human emotions not as isolated categories, but as points within a continuous two-dimensional space[cite: 1].

Any emotion can be mapped along two axes[cite: 1]:

- **Valence (Horizontal Axis):** Measures the intrinsic pleasantness of an emotion, ranging from negative/unpleasant (left) to positive/pleasant (right)[cite: 1].
- **Activation / Arousal (Vertical Axis):** Measures the energy level or physical intensity of the emotion, ranging from low/calm (bottom) to high/excited (top)[cite: 1].

Crossing these axes yields four distinct quadrants (4Q), which serve as the ground-truth labels in EMOPIA[cite: 1]:

Quadrant	Valence	Activation (Arousal)	Associated Emotions
Q1	High	High	Joy, excitement, triumph
Q2	Low	High	Anger, tension, anxiety
Q3	Low	Low	Sadness, depression, melancholy
Q4	High	Low	Calmness, tenderness, relaxation

Table 1: The four quadrants of Russell’s circumplex model[cite: 1].

Analyzing the MIDI files confirms that placement within Russell’s four quadrants correlates strongly with fundamental musical attributes¹[cite: 1]:

- **Relationship with Arousal:** Emotional intensity directly correlates with performance energy[cite: 1]. High-arousal quadrants (Q1 and Q2) display high note density (more notes per beat) and high velocity (stronger key attacks)[cite: 1]. Conversely, low-arousal quadrants (Q3 and Q4) feature sustained notes played with softer velocity[cite: 1].
- **Relationship with Valence:** Emotional positivity is tied to harmonic tonality[cite: 1]. Key distributions show that high-valence pieces (Q1 and Q4) predominantly use major scales[cite: 1]. In contrast, low-valence pieces (Q2 and Q3), which convey tension or sadness, heavily rely on minor scales[cite: 1].

2.2 MIDI Data Tokenization

The raw dataset comprises 1,078 MIDI sequences[cite: 1]. MIDI files represent note events, key velocities, and durations with precision, eliminating audio background noise[cite: 1]. However, raw MIDI files cannot be directly fed into neural networks[cite: 1]. To model music using Natural Language Processing (NLP) techniques, the data must be tokenized[cite: 1].

We use the `MidiTok` library² to convert MIDI files into sequences of musical tokens via the REMI (REvamped MIDI) representation³[cite: 1].

This method converts continuous musical events into discrete categorical variables (tokens)[cite: 1]. REMI aligns music along a metric grid (bars and sub-beats), effectively capturing tempo and rhythm[cite: 1]. Notes are contextualized by their positional placement within a bar[cite: 1]. A piece of music is thus represented as a sequence of structured event tokens[cite: 1]:

1. **Bar:** Indicates the start of a new measure[cite: 1].
2. **Position:** Subdivides the measure along a grid (e.g., 16 or 32 steps) to specify note onset[cite: 1].
3. **Pitch:** Represents note height (MIDI pitch range 0–127)[cite: 1].
4. **Velocity:** Represents key strike force (quantized to 16 levels for training efficiency)[cite: 1].
5. **Duration:** Specifies note length in fractions of a beat[cite: 1].

For instance, a single note event translates to: `[Bar_0] → [Position_0] → [Pitch_60] → [Velocity_12] → [Duration_2.0] → [Position_8] → ...`[cite: 1]

Our final vocabulary size consists of 268 unique tokens[cite: 1].

2.3 Statistical Analysis and Initial Separability

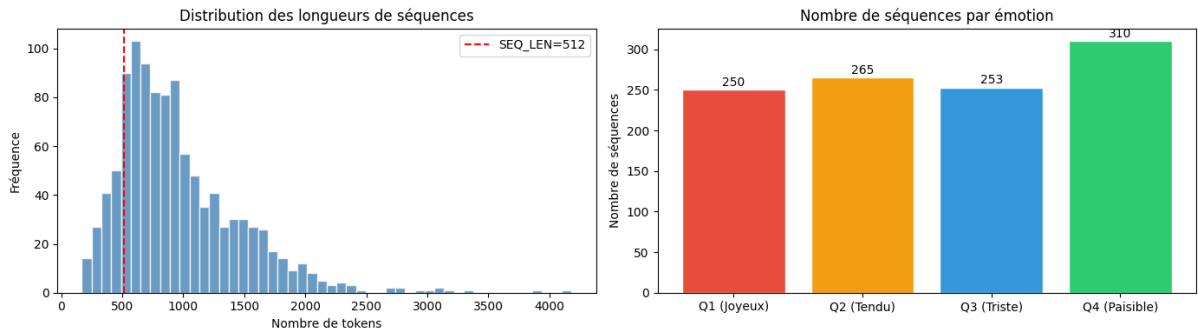


Figure 1: Exploratory data analysis of the EMOPIA dataset[cite: 1].

To understand the dataset structure prior to model training, we performed an exploratory analysis on the tokenized sequences[cite: 1].

As shown in the class distribution bar plot (right), the dataset is relatively balanced across the four emotions, ranging from 250 samples in Q1 (Joyful) to 310 samples in Q4 (Peaceful)[cite: 1]. This balance prevents the model from developing significant statistical bias toward any single emotion during training[cite: 1].

Regarding sequence length, the histogram (left) shows that most original sequences contain between 400 and 1,500 tokens[cite: 1]. Because neural network batch training requires uniform tensor dimensions, sequence length normalization is necessary[cite: 1].

We fixed sequence lengths via truncation, setting a maximum length of `SEQ_LEN = 512` tokens based on empirical testing[cite: 1]. While truncation sacrifices some late-stage musical development, 512 tokens preserve a substantial portion of the original composition while maintaining training efficiency[cite: 1]. Shorter sequences are padded with zeros (`padding_idx = 0`)[cite: 1].

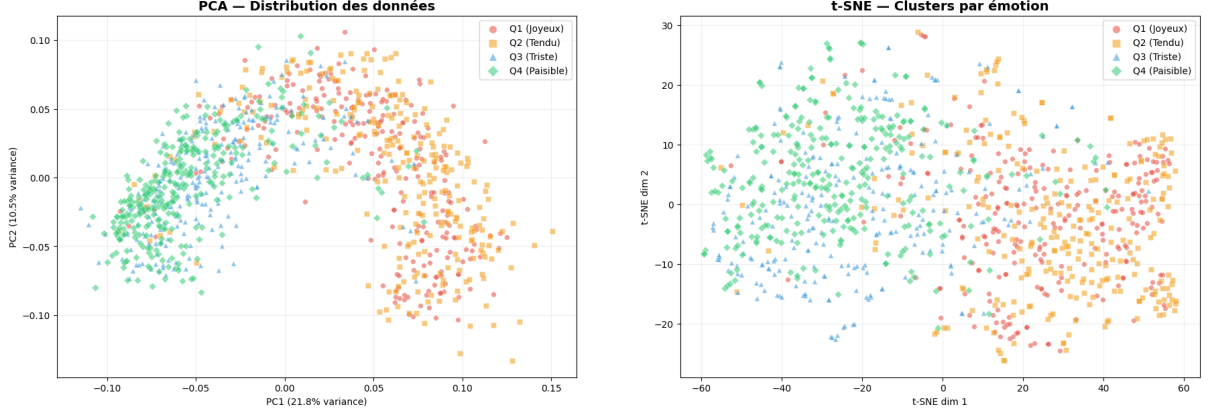


Figure 2: Dimensionality reduction plots of the raw input data[cite: 1].

2.4 t-SNE Projection

To visualize high-dimensional token distributions, we employ t-SNE⁴[cite: 1]. t-SNE preserves local topology, mapping similar high-dimensional musical sequences to nearby locations in 2D space[cite: 1].

The t-SNE algorithm converts Euclidean distances between points into conditional probabilities[cite: 1]. The probability $p_{j|i}$ that point x_i picks x_j as its neighbor is proportional to a Gaussian density centered at x_i [cite: 1]:

$$p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2 / 2\sigma_i^2)}$$

In the low-dimensional space (y_i), neighborhood similarities q_{ij} are modeled using a Student-t distribution with one degree of freedom (Cauchy distribution) to prevent crowding[cite: 1]:

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq i} (1 + \|y_i - y_k\|^2)^{-1}}$$

The algorithm minimizes the Kullback-Leibler (KL) divergence between p and q via gradient descent[cite: 1]:

$$C = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

Interpretation of Raw Data t-SNE:

Projecting the raw token frequencies into 2D yields a heavily overlapped distribution across all four quadrants[cite: 1]. No clear natural boundaries separate joy, sadness, calmness, and tension[cite: 1]. We will use t-SNE as a benchmark to evaluate model performance: if a model successfully forms well-separated clusters corresponding to emotional labels, it demonstrates mathematically that the deep architecture has structured the latent space around emotional attributes[cite: 1].

2.5 Principal Component Analysis (PCA)

We also apply Principal Component Analysis (PCA),⁵ a linear technique that seeks orthogonal axes maximizing data variance[cite: 1]. Here, each track is represented by the occurrence frequency of its 268 tokens[cite: 1]. PCA computes the eigenvectors and eigenvalues of the covariance matrix[cite: 1]. The first principal component (PC1, horizontal axis) captures 21.8% of total dataset variance, while the second (PC2, vertical axis) captures 10.5%[cite: 1]. Together, the 2D projection accounts for 32.3% of the total variance[cite: 1]. Visually, the projected points form a distinct horseshoe curve—a phenomenon known as the Guttman effect[cite: 1]. This curvature indicates that musical pieces vary along a continuous progression rather than discrete clusters[cite: 1]. However, emotional categories overlap completely along this curve, demonstrating a total lack of linear separability[cite: 1].

This linear overlap justifies the need for deep learning[cite: 1]. Because linear methods like PCA fail to separate musical emotions, non-linear deep generative architectures—such as Conditional Variational Autoencoders (CVAE)—are necessary to isolate emotional representations[cite: 1].

3 The Variational Autoencoder (VAE)

We first evaluate standard VAE architectures to establish an effective baseline for musical sequence reconstruction before introducing conditional mechanisms[cite: 1].

3.1 Embedding Layer

To process discrete tokens through neural networks, sequences are mapped into a continuous vector space using an embedding layer (`nn.Embedding`)[cite: 1].

The embedding layer acts as a lookup matrix $W_E \in \mathbb{R}^{|V| \times d}$, where $|V| = 268$ is the vocabulary size and $d = 128$ is the embedding dimension[cite: 1]. Discrete token indices are mapped to dense continuous vectors, whose weights are optimized end-to-end via backpropagation to group semantically similar musical events together in vector space[cite: 1].

3.2 VAE Architecture

A Variational Autoencoder (VAE) compresses input data into a lower-dimensional latent space before reconstructing it[cite: 1]. Unlike standard autoencoders, a VAE enforces a regularized probabilistic distribution over the latent space[cite: 1].

Applying autoencoders to symbolic music is supported by state-of-the-art audio synthesis literature⁶[cite: 1]. Music exhibits complex temporal dependencies; compressing sequence representations into a structured latent space forces the network to extract global features such as tonality, rhythm, and expressiveness[cite: 1].

The overall VAE pipeline is structured as follows[cite: 1]:

$$x \text{ (MIDI tokens)} \rightarrow \text{Encoder} \rightarrow \mu, \log(\sigma^2) \rightarrow z \sim \mathcal{N}(\mu, \sigma^2) \rightarrow \text{Decoder} \rightarrow \hat{x} \text{ (reconstruction)}$$

3.2.1 The Reparameterization Trick

Direct stochastic sampling $z \sim \mathcal{N}(\mu, \sigma^2)$ blocks gradient backpropagation[cite: 1]. The reparameterization trick resolves this by isolating the random variable ϵ [cite: 1]:

$$z = \mu + \sigma \cdot \epsilon \quad \text{where} \quad \epsilon \sim \mathcal{N}(0, 1)$$

Because ϵ is sampled independently, gradients flow directly through μ and σ to update encoder weights[cite: 1].

3.2.2 Loss Function: Evidence Lower Bound (ELBO)

VAEs maximize the Evidence Lower Bound (ELBO)[cite: 1]:

$$\text{ELBO} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - \text{KL}(q(z|x) || p(z))$$

During training, we minimize the negative ELBO (Loss = Reconstruction Loss + $\beta \cdot \text{KL Loss}$)[cite: 1].

1. Reconstruction Loss: Measures how accurately the decoder reconstructs the input sequence using Cross-Entropy Loss[cite: 1]:

$$\mathcal{L}_{recon} = - \sum_{t=1}^T \log p(x_t|z)$$

Padding tokens (index = 0) are masked using `ignore_index=0` to prevent gradient distortion[cite: 1].

2. KL Divergence: Penalizes deviations between the learned latent distribution $q(z|x)$ and the standard Gaussian prior $p(z) = \mathcal{N}(0, I)$ [cite: 1]:

$$\mathcal{L}_{KL} = \text{KL}(q||p) = -\frac{1}{2} \sum_{j=1}^d (1 + \log(\sigma_j^2) - \mu_j^2 - \sigma_j^2)$$

KL Annealing (β -warmup): To mitigate posterior collapse—where the decoder ignores latent variable z early in training—we weight the KL term by a factor β that gradually increases from 0 to 1 across initial epochs[cite: 1].

3.2.3 Target Training Dynamics

A well-optimized model should display three distinct behaviors[cite: 1]:

1. Reconstruction loss drops steadily as the network learns sequence structure[cite: 1].
2. KL divergence increases initially before stabilizing, reflecting latent space organization[cite: 1].
3. Train and validation losses converge symmetrically, confirming good generalization[cite: 1].

3.3 Training Setup and Regularization

- **Data Splitting:** The dataset is split into training and validation sets to monitor generalization and prevent overfitting[cite: 1].
- **Optimization Objective:** Models are trained iteratively to minimize the negative ELBO loss over the validation set[cite: 1].
- **Early Stopping:** Training halts automatically if validation loss fails to improve over a set patience window, saving the best-performing checkpoint[cite: 1].
- **Model Selection:** Final evaluation is conducted using the model checkpoint that achieved the lowest validation loss[cite: 1].

3.4 Test 1: Baseline MLP-VAE

We first evaluated a Multi-Layer Perceptron (MLP) VAE baseline to validate our data pipeline—from raw MIDI tokenization to generated MIDI conversion—before moving to complex architectures[cite: 1].

- **Configuration:** The encoder flattened sequence embeddings ($256 \times 64 = 16,384$ dimensions) and passed them through linear layers (512 and 256 units) to predict μ and $\log(\sigma^2)$ in a 32-dimensional latent space[cite: 1]. The decoder mirrored this setup to reconstruct output logits (256×268)[cite: 1]. The architecture contained over 87 million parameters[cite: 1].
- **Generation Sampling Temperature:** Latent vectors z were sampled directly from $\mathcal{N}(0, I)$ [cite: 1]. Logits were converted to tokens using two decoding modes[cite: 1]:
 - *Deterministic Mode* ($Temp \leq 0$): Argmax selection of the highest-probability token[cite: 1].
 - *Stochastic Mode* ($Temp > 0$): Softmax sampling weighted by temperature[cite: 1]. Lower temperatures produce conservative, repetitive sequences, while higher temperatures introduce variety at the cost of potential harmonic dissonance[cite: 1].

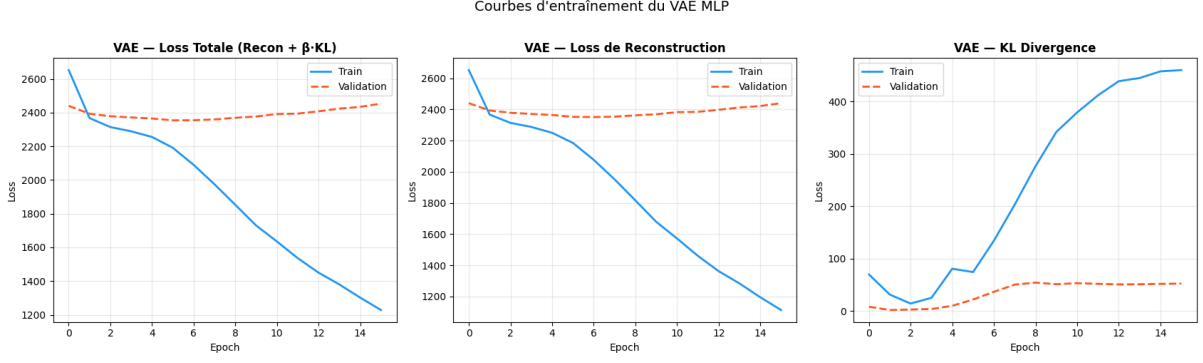


Figure 3: Training and validation loss curves for the MLP-VAE baseline[cite: 1].

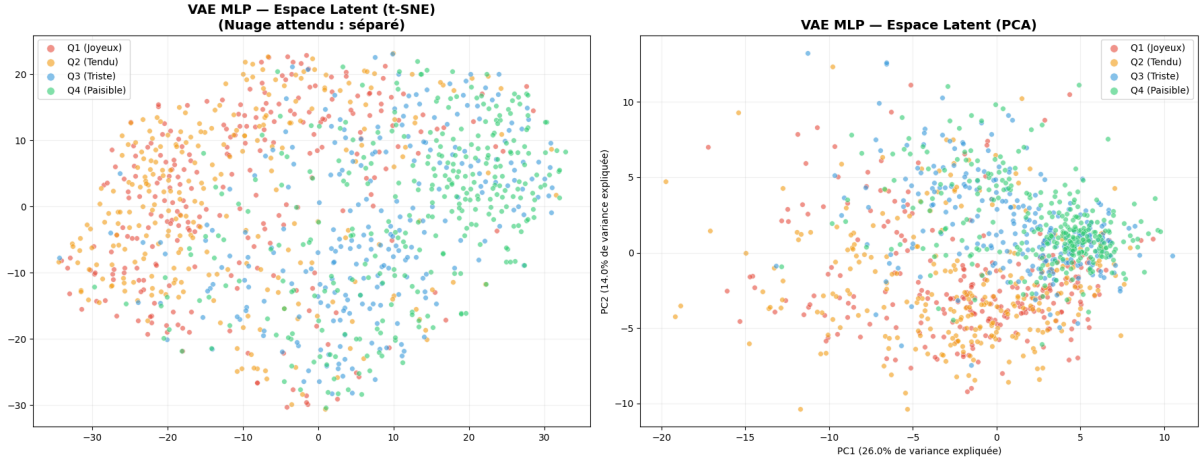


Figure 4: Latent space visualisations (t-SNE and PCA) for the MLP-VAE[cite: 1].

- **Results and Failure Modes:** The model suffered from severe overfitting: training loss dropped to 1,200 while validation loss stalled at 2,500[cite: 1]. Uncontrolled KL divergence growth indicated that the encoder scattered training points across latent space to memorize inputs rather than learning generalizable structures[cite: 1]. Latent space plots (t-SNE/PCA) showed no clear emotional clustering[cite: 1]. However, a binary separation emerged along the arousal axis (high-energy Q1/Q2 vs. low-energy Q3/Q4), indicating that while arousal was coarsely captured, valence remained completely unresolvable[cite: 1].
- **Musical Quality Limitations:** Output music was erratic, short, and unmusical[cite: 1]. By flattening input sequences, the MLP destroyed sequential ordering, rendering the model incapable of learning rhythm, continuous melody, or harmonic context[cite: 1].

3.5 Test 2: Spatial Approach (1D-CNN VAE)

To address the parameter explosion and lack of spatial awareness in MLPs, we implemented a 1D Convolutional Neural Network (1D-CNN) VAE to capture localized patterns such as chords and rhythmic motives[cite: 1].

- **Configuration:** The encoder used three 1D convolutional layers with ReLU activations and Dropout[cite: 1]. The decoder used 1D transposed convolutions to reconstruct sequence dimensions[cite: 1]. Total parameter count dropped drastically to 539,000[cite: 1].
- **Results:** Train and validation losses aligned tightly, eliminating massive overfitting[cite: 1]. Reconstruction loss improved to 1,710[cite: 1]. However, KL divergence collapsed to near zero (116),

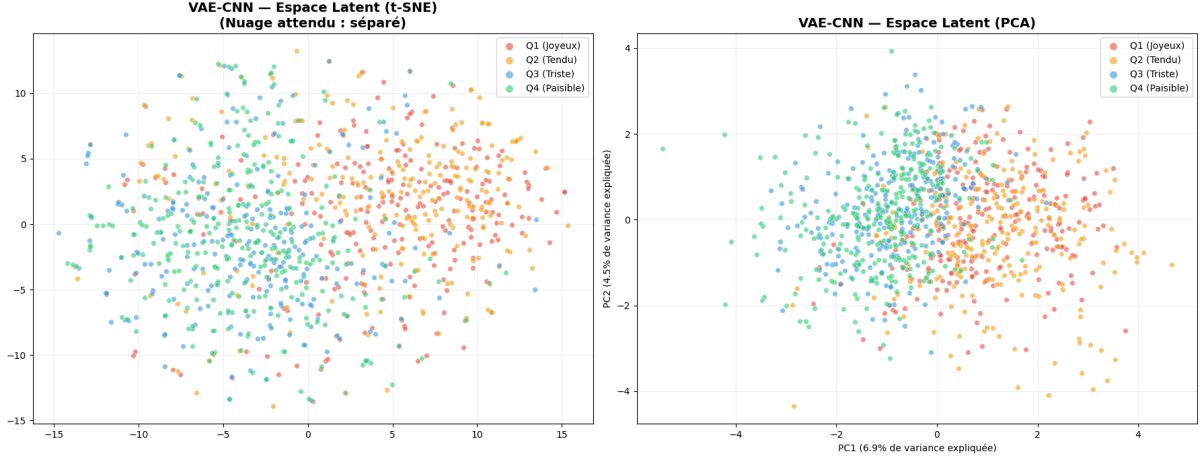


Figure 5: Loss curves for the 1D-CNN VAE model[cite: 1].

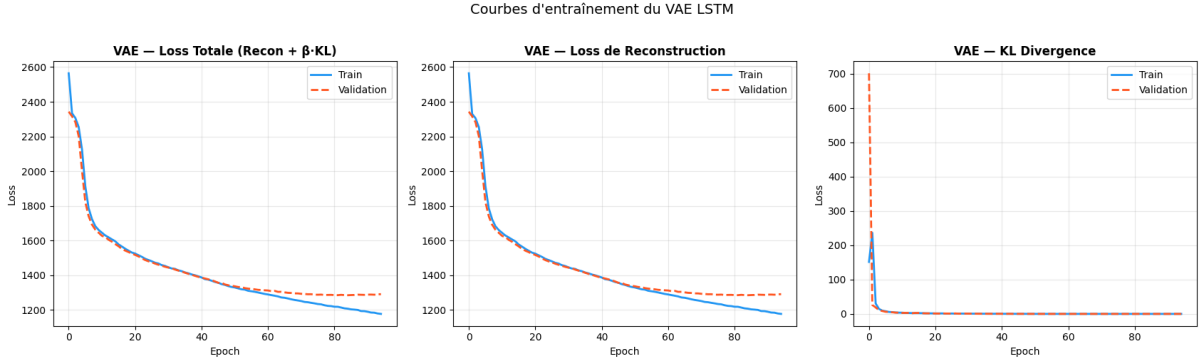


Figure 6: Latent space visualization for the 1D-CNN VAE[cite: 1].

signaling posterior collapse: the decoder ignored z and acted as a standard autoencoder[cite: 1]. Latent projections showed no emotional separation[cite: 1].

- **Musical Quality Limitations:** Generated audio sounded fractured and filled with random silences[cite: 1]. Because CNNs generate tokens in parallel across receptive fields, they failed to enforce strict REMI grammatical ordering (**Bar** → **Position** → **Pitch** → **Velocity** → **Duration**) and lacked the long-term context required for sustained melodic cohesion[cite: 1].

3.6 Test 3: Sequential Approach (LSTM-VAE)

To model long-range temporal dependencies and sequential syntax, we implemented a Long Short-Term Memory (LSTM) VAE[cite: 1].

3.6.1 The LSTM Architecture

LSTMs mitigate vanishing gradients in recurrent networks by maintaining a memory cell c_t alongside hidden state h_t [cite: 1]. Information flow is regulated by three gating mechanisms[cite: 1]:

1. **Forget Gate:** Discards irrelevant past information from c_{t-1} [cite: 1].
2. **Input Gate:** Writes relevant new information from token x_t to c_t [cite: 1].
3. **Output Gate:** Filters cell state c_t to update hidden state h_t [cite: 1].

This architecture allows the network to preserve key musical structures—such as key signatures, themes, and meter—across extended sequences[cite: 1].

- **Model Setup:** The encoder comprised a bidirectional LSTM (capturing forward and backward context), while the decoder used an autoregressive unidirectional LSTM[cite: 1].
- **Shifted Teacher Forcing:** To prevent the decoder from trivial copy-pasting during training, target inputs were shifted by one timestep ($t - 1$), forcing the model to predict step $t + 1$ based strictly on prior context[cite: 1].
- **Results:** Validation loss dropped to 1,284.4 (outperforming CNN’s 2,100 and MLP’s 2,400)[cite: 1]. The latent space exhibited distinct, highly localized micro-clusters[cite: 1]. However, these clusters were multicolored, containing points from all four quadrants[cite: 1]. This indicated that while the encoder organized the latent space around musical structural traits, these traits did not map to emotional categories[cite: 1].
- **Musical Assessment:** Generated samples were coherent, rhythmic, and musically structured[cite: 1]. Fast-tempo major-key structures emerged in Q1-prompted samples, while slower minor-key structures appeared in Q3-prompted samples[cite: 1]. Nevertheless, because emotional attributes were entangled implicitly within the latent space, generation could not be explicitly controlled[cite: 1].

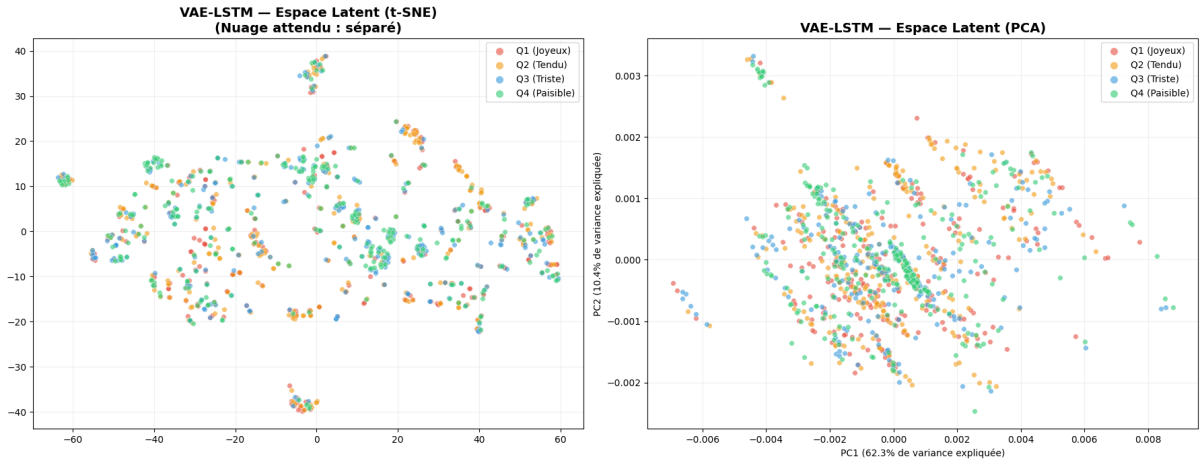


Figure 7: t-SNE latent projection of the unconditioned LSTM-VAE ($d = 64$)[cite: 1].

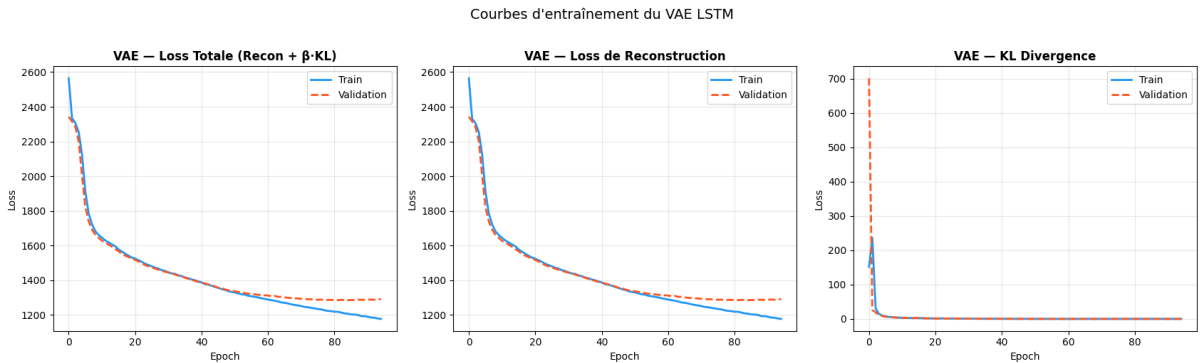


Figure 8: Loss curves for the LSTM-VAE ($d = 64$)[cite: 1].

While the LSTM-VAE generated realistic music, generation remained unguided[cite: 1]. Controlling output emotion required transitioning to a conditional framework: the CVAE[cite: 1].

4 The Conditional VAE (CVAE)

The Conditional Variational Autoencoder (CVAE) extends the standard VAE by conditioning encoder and decoder distributions on explicit attribute vectors c ⁷[cite: 1].

In our system, c represents the emotion quadrant (Q1–Q4) encoded as a 4D one-hot vector (e.g., $[1, 0, 0, 0]$ for Q1)[cite: 1].

Standard VAE distributions[cite: 1]:

$$\text{Encoder: } q(z|x) \quad \text{and} \quad \text{Decoder: } p(x|z)$$

CVAE conditional distributions⁸[cite: 1]:

$$\text{Encoder: } q(z|x, c) \quad \text{and} \quad \text{Decoder: } p(x|z, c)$$

The operational pipeline becomes[cite: 1]:

$$x, c \rightarrow \text{Encoder} \rightarrow \mu, \log(\sigma^2) \rightarrow z \sim \mathcal{N}(\mu, \sigma^2) \rightarrow [z, c] \rightarrow \text{Decoder} \rightarrow \hat{x}$$

- **Encoder Conditioning:** c is concatenated with input sequence embeddings at each timestep[cite: 1]. Because the encoder knows the target emotion via c , z is freed from encoding class labels and instead focuses purely on style, rhythm, and fine-grained variations[cite: 1].
- **Decoder Conditioning:** During inference, a style vector $z \sim \mathcal{N}(0, I)$ is concatenated with a target emotion vector c [cite: 1]. The combined representation $[z, c]$ guides autoregressive decoding, forcing velocity, tempo, and harmony to align with the specified emotion[cite: 1].

5 Generation and Results

Inference sampling uses a temperature-scaled Softmax to balance musical coherence and creative variation[cite: 1]. Extremely low temperatures cause repetitive looping, while high temperatures produce harmonic errors[cite: 1]. Empirical testing identified Temperature = 0.6 as optimal for classical/pop piano generation[cite: 1].

5.1 Convergence and Latent Emotion Separation

We evaluated two latent space dimensions ($d = 32$ vs. $d = 64$)[cite: 1].

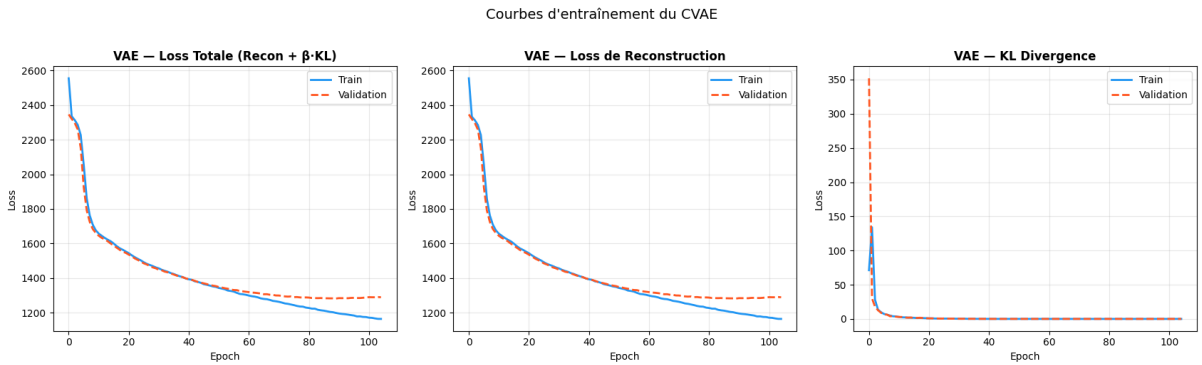


Figure 9: Training curves for CVAE-LSTM ($d = 32$)[cite: 1].

1. Loss Dynamics: Both configurations trained stably without overfitting, with validation losses dropping from 2,600 to 1,180[cite: 1]. Initial KL divergence peaked higher for $d = 64$ (750) than $d = 32$ (350), reflecting the higher capacity of the 64D space[cite: 1].

2. Latent Space Disentanglement (PCA t-SNE): In contrast to raw data and standard VAE projections, the CVAE latent space formed four completely isolated clusters corresponding to Russell’s quadrants[cite: 1].

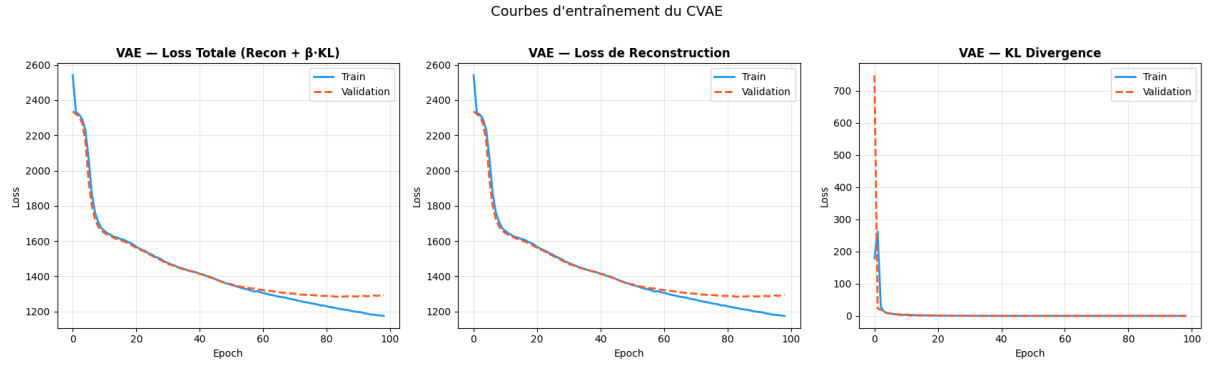


Figure 10: Training curves for CVAE-LSTM ($d = 64$)[cite: 1].

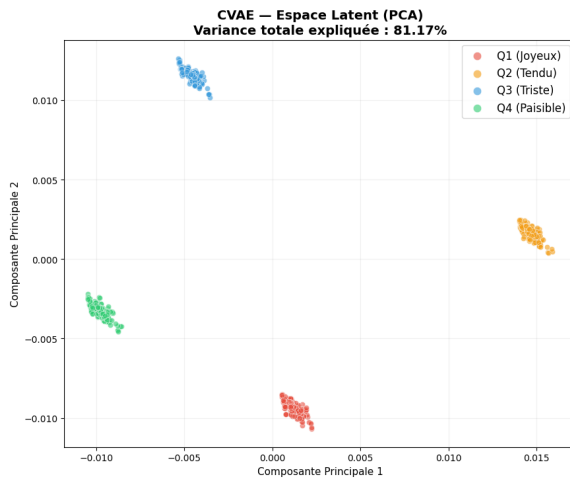


Figure 11: PCA ($d = 32$)[cite: 1].

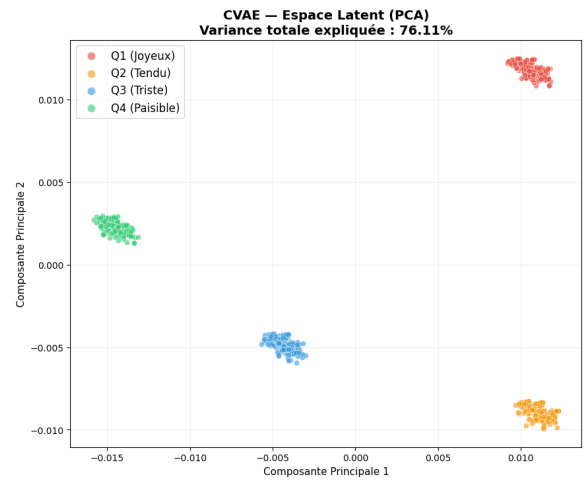


Figure 12: PCA ($d = 64$)[cite: 1].

- **PCA Analysis:** PCA explained 81.17% of variance for $d = 32$ and 76.11% for $d = 64$, cleanly separating Q1 (red), Q2 (orange), Q3 (blue), and Q4 (green)[cite: 1].

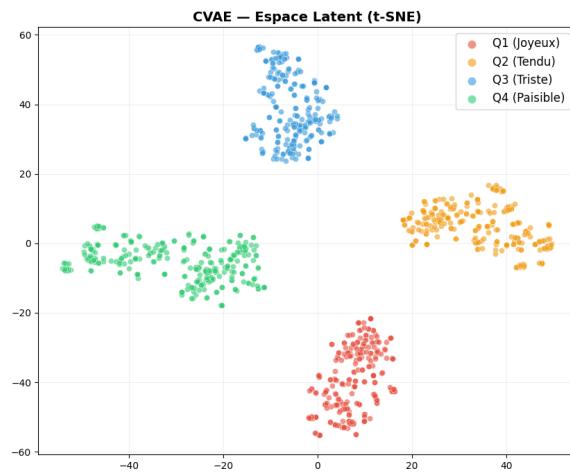


Figure 13: t-SNE ($d = 32$)[cite: 1].

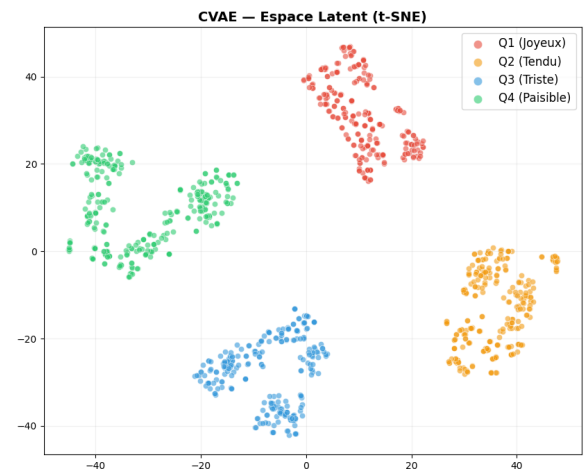


Figure 14: t-SNE ($d = 64$)[cite: 1].

- **t-SNE Analysis:** $d = 32$ yielded tightly packed clusters, whereas $d = 64$ produced broader, well-separated clusters[cite: 1]. The increased variance in $d = 64$ provides the decoder with greater stylistic flexibility within each emotion category[cite: 1].

3. Qualitative Listening Evaluation: Forcing condition c successfully controlled output characteristics[cite: 1]. Q1/Q2 prompts generated high-velocity, rapid-tempo passages, whereas Q4 prompts yielded calm, sparse textures[cite: 1]. Audio quality was noticeably richer and more natural for $d = 64$ compared to $d = 32$ [cite: 1].

6 Conclusion and Future Work

While the CVAE-LSTM model generates convincingly textured emotional compositions, maintaining long-term macro-structure (e.g., repeating themes across several minutes) remains a key challenge for LSTM backbones[cite: 1].

6.1 Model Performance Summary

Validation performances across evaluated architectures are summarized below[cite: 1]:

Model	Latent Dim	Final Val Loss	Key Observation
VAE (MLP)	64	2346.0	Severe overfitting, KL collapse, no temporal structure
VAE (CNN)	64	1885.4	Stable convergence, temporal amnesia, unnatural pauses
VAE (LSTM)	64	1284.4	Strong musical grammar, unguided emotional generation
C-VAE (LSTM)	64	1283.6	Best audio quality, clear 4-cluster emotional separation

Table 2: Validation performance comparison across architectures[cite: 1].

Performance Analysis: The performance progression highlights key architectural insights[cite: 1]. The MLP failed due to sequence flattening (2346.0)[cite: 1]. 1D-CNNs improved performance (1885.4) by capturing local patterns, but lacked sequence memory[cite: 1]. Recurrent LSTMs achieved a major breakthrough (1284.4), proving that musical modeling requires sequential memory[cite: 1]. Finally, the conditional model (CVAE) maintained this low validation loss (1283.6) while successfully enforcing explicit emotional control[cite: 1].

Future extensions include replacing LSTMs with Music Transformers to leverage self-attention for long-range structural coherence,³ or incorporating Generative Adversarial Networks (GANs)⁶ to enhance stylistic realism[cite: 1].

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